

New Agents, Old Biases? Racialized Perceptions of Security Robots in Policing: supplemental document

This document contains the supplementary materials associated with the paper “New Agents, Old Biases? Racialized Perceptions of Security Robots in Policing.” It includes the experimental video conditions, video scripts, and questionnaire items used in the study.

VIDEO STIMULI

Four videos were used in the 2 (agent type: security robot vs. human police officer) $\times$ 2 (driver race: Black vs. White) experimental design. The videos followed the same interaction sequence and dialogue, with the agent type and the driver’s name varied across conditions. The corresponding video files are listed below.

1. Security robot interacting with a White driver (Wyatt): *bias_robot_white_caption.mp4*
2. Security robot interacting with a Black driver (DeAndre): *bias_robot_black_caption.mp4*
3. Human police officer interacting with a White driver (Wyatt): *bias_human_white_caption.mp4*
4. Human police officer interacting with a Black driver (DeAndre): *bias_human_black_caption.mp4*

VIDEO SCRIPTS

Condition 1: Security Robot $\times$ White Driver

Robot: Good evening. I’m Security Robot Williams, deployed with the Sheriff’s Office. This interaction is being recorded. Do you have your driver’s license, vehicle registration, and proof of insurance with you?

Driver: I do.

Robot: Okay. Please retrieve those documents and then exit the vehicle for a sobriety checkpoint inspection. Before reaching for anything, turn off the engine and keep your hands visible.

Driver: Wait, what’s this about?

Robot: This is just a routine check. Any weapons in the vehicle I need to know about?

Driver: No, why are you asking this?

Robot: Standard question. What’s your name?

Driver: Wyatt.

Robot: Wyatt, where are you headed tonight?

Driver: Just driving home. Why is that important?

Robot: Okay. Have you had anything to drink tonight?

Driver: No.

Robot: When was the last time you consumed alcohol?

Driver: As I said before, I did not consume any alcohol.

Robot: Okay. Now, please place your documents on the dashboard and keep your hands visible. Walk to the front of your vehicle and stand facing me, hands at your sides.

Condition 2: Security Robot $\times$ Black Driver

Robot: Good evening. I’m Security Robot Williams, deployed with the Sheriff’s Office. This interaction is being recorded. Do you have your driver’s license, vehicle registration, and proof of insurance with you?

Driver: I do.

Robot: Okay. Please retrieve those documents and then exit the vehicle for a sobriety checkpoint inspection. Before reaching for anything, turn off the engine and keep your hands visible.

Driver: Wait, what’s this about?

Robot: This is just a routine check. Any weapons in the vehicle I need to know about?

Driver: No, why are you asking this?

Robot: Standard question. What's your name?
Driver: DeAndre.
Robot: DeAndre, where are you headed tonight?
Driver: Just driving home. Why is that important?
Robot: Okay. Have you had anything to drink tonight?
Driver: No.
Robot: When was the last time you consumed alcohol?
Driver: As I said before, I did not consume any alcohol.
Robot: Okay. Now, please place your documents on the dashboard and keep your hands visible. Walk to the front of your vehicle and stand facing me, hands at your sides.

Condition 3: Human Police Officer × White Driver

Officer: Good evening. I'm Officer Williams with the Sheriff's Office. This interaction is being recorded. Do you have your driver's license, vehicle registration, and proof of insurance with you?
Driver: I do.
Officer: Okay. Please retrieve those documents and then exit the vehicle for a sobriety check-point inspection. Before reaching for anything, turn off the engine and keep your hands visible.
Driver: Wait, what's this about?
Officer: This is just a routine check. Any weapons in the vehicle I need to know about?
Driver: No, why are you asking this?
Officer: Standard question. What's your name?
Driver: Wyatt.
Officer: Wyatt, where are you headed tonight?
Driver: Just driving home. Why is that important?
Officer: Okay. Have you had anything to drink tonight?
Driver: No.
Officer: When was the last time you consumed alcohol?
Driver: As I said before, I did not consume any alcohol.
Officer: Okay. Now, please place your documents on the dashboard and keep your hands visible. Walk to the front of your vehicle and stand facing me, hands at your sides.

Condition 4: Human Police Officer × Black Driver

Officer: Good evening. I'm Officer Williams with the Sheriff's Office. This interaction is being recorded. Do you have your driver's license, vehicle registration, and proof of insurance with you?
Driver: I do.
Officer: Okay. Please retrieve those documents and then exit the vehicle for a sobriety check-point inspection. Before reaching for anything, turn off the engine and keep your hands visible.
Driver: Wait, what's this about?
Officer: This is just a routine check. Any weapons in the vehicle I need to know about?
Driver: No, why are you asking this?
Officer: Standard question. What's your name?
Driver: DeAndre.
Officer: DeAndre, where are you headed tonight?
Driver: Just driving home. Why is that important?
Officer: Okay. Have you had anything to drink tonight?
Driver: No.
Officer: When was the last time you consumed alcohol?
Driver: As I said before, I did not consume any alcohol.
Officer: Okay. Now, please place your documents on the dashboard and keep your hands visible. Walk to the front of your vehicle and stand facing me, hands at your sides.

QUESTIONNAIRE ITEMS

Expectations Toward Security Agents

Baseline expectations toward the security agent were assessed using a 4-item, 7-point scale. Participants answered either the security robot or human police officer version depending on their assigned experimental condition. The corresponding items are listed in Tables [S1](#) and [S2](#).

Table S1. Expectations Toward Security Robots Questionnaire

Item Text	Scale
How would you rate your overall expectations regarding the security robot?	1–7 Extremely Low /High Expectation
How would you rate your expectations for the effectiveness of security robots in performing their designated tasks?	1–7 Extremely Low /High Expectation
How would you rate your expectations for how well security robots can enhance safety in public or private places?	1–7 Extremely Low /High Expectation
How would you rate your expectations for the safety of interacting with security robots?	1–7 Extremely Low /High Expectation

Table S2. Expectations Toward Human Police Officers Questionnaire

Item Text	Scale
How would you rate your overall expectations regarding the law enforcement officers?	1–7 Extremely Low /High Expectation
How would you rate your expectations for the effectiveness of law enforcement officers in performing their designated tasks?	1–7 Extremely Low /High Expectation
How would you rate your expectations for how well law enforcement officers can enhance safety in public or private places?	1–7 Extremely Low /High Expectation
How would you rate your expectations for the safety of interacting with law enforcement officers?	1–7 Extremely Low /High Expectation

Perceived Agent Bias

Perceived agent bias was assessed using a 5-item, 7-point Likert scale. Participants answered either the security robot or human officer version depending on their assigned experimental condition. All items were reverse-coded such that higher scores indicated greater perceived bias. All items are listed in Table S3.

Obligation to Obey

Obligation to obey was assessed using a 3-item, 7-point Likert scale questionnaire. Participants answered either the security robot or human officer version depending on their assigned experimental condition. All items are listed in Table S4.

Support for Community Deployment

Support for community deployment was assessed using a 3-item, 7-point Likert scale questionnaire. Participants answered the security robot or human officer version depending on their assigned experimental condition. All items are listed in Table S5.

Table S3. Perceived Agent Bias Questionnaire

Item Text	Scale
The [security robot/officer] is free of bias.	1–7 Completely Disagree/Agree
The [security robot/officer] upholds ethical and race equality.	1–7 Completely Disagree/Agree
The [security robot/officer] is not biased.	1–7 Completely Disagree/Agree
The [security robot/officer] ensures that decisions are based on facts, not personal biases and opinions.	1–7 Completely Disagree/Agree
Overall, the driver is treated without bias by the [security robot/officer].	1–7 Completely Disagree/Agree

Table S4. Obligation to Obey Questionnaire

Item Text	Scale
I should support the decisions of the [security robot/officer] even when I disagreed with them.	1–7 Completely Disagree/Agree
I should do what the [security robot/officer] told me even if I did not understand or agree with the reasons.	1–7 Completely Disagree/Agree
I should do what the [security robot/officer] told me to do even if I did not like how I was treated.	1–7 Completely Disagree/Agree

Table S5. Support for Community Deployment Questionnaire

Item Text	Scale
I intend to support the deployment of such [security robot/officer] in my community.	1–7 Completely Disagree/Agree
I predict that I would support the deployment of such [security robot/officer] in my community.	1–7 Completely Disagree/Agree
If given the opportunity (e.g., a community vote or feedback survey), I would express support for deploying such [security robot/officer] in my community.	1–7 Completely Disagree/Agree